

Raman

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SKILLS

Languages:	C++, Python, C, Java, SQL
AI/ML:	Machine Learning, Natural Language Processing, NumPy, Pandas, Matplotlib
Tools/Platforms:	Git, GitHub, Jupyter Notebook, Blender
Soft Skills:	Problem-Solving, Adaptability, Teamwork

PROJECTS

CodeExplain <i>Python, Streamlit, Tree Sitter</i>	July' 26
- Developed a plain-English code tutoring application that helps beginners understand unfamiliar code through automated explanations, complexity analysis, algorithm identification, and interactive learning. - Implemented AST-based static code analysis using Tree-sitter to extract functions, classes, loops, conditionals, variables, imports, and function calls. - Integrated Groq LLaMA 3.3 70B with prompt engineering and structured JSON generation to produce line-by-line explanations, complexity analysis, and automated quizzes. - GitHub repository. View	
Deadlock Prevention and Recovery Kit <i>HTML5, CSS3, JavaScript, Banker's Algorithm</i>	Mar' 26
- Developed an interactive Operating Systems simulator that visually demonstrates deadlock avoidance, detection, and recovery through configurable process and resource scenarios. - Implemented Banker's Algorithm with dynamic Allocation, Maximum, Available, and Need matrices to identify safe/unsafe system states and generate safe execution sequences. - Built deadlock detection and recovery mechanisms with process termination, resource preemption, rollback simulation, and Canvas-based Resource Allocation Graph visualization. - GitHub repository. View	
AI Nutritionist Chatbot <i>Python, RAG, LLM, FAISS, FastAPI, Streamlit, Groq</i>	May' 26
- Developed a knowledge-grounded conversational assistant that provides personalized nutrition guidance, diet recommendations, and calorie/macro insights through natural-language queries. - Implemented a FAISS-based RAG pipeline with Sentence Transformers, vector embeddings, similarity search, and SHA-256-based caching for efficient knowledge retrieval. - Integrated LLM inference and prompt engineering with domain-safety controls to generate structured responses with explanations, recommendations, nutritional breakdowns, and safety notes. - GitHub repository. View	

TRAINING

Mastering LLMs And Agentic AI <i>LPU</i> Link	
- Completed training in LLMs, Transformers, Prompt Engineering, RAG, Vector Databases, Fine-Tuning, and AI Agents. - Developed Code Explain, an AI-powered code tutoring application for automated code explanation, complexity analysis, algorithm identification, and interactive quizzes.	
	Jul' 26

CERTIFICATES

Basic to Beyond Python <i>CSE Pathshala</i> Link	Jan' 25
Data Structure and Algorithms <i>iamneo</i> Link	Jun' 26
Full Stack Web Developer Workshop <i>Physics Wallah</i> Link	Nov' 24
React JS <i>Tech Veda</i> Link	Mar' 25

EDUCATION

Lovely Professional University	Phagwara, Punjab
Bachelor of Technology	Aug' 24 – Present
Computer Science and Engineering; CGPA: 7.43	
Suditi Global Academy	Etawah, Uttar Pradesh
Intermediate	April 23 – March 24
Percentage: 74%	